WALL-E shield: the last few measurements

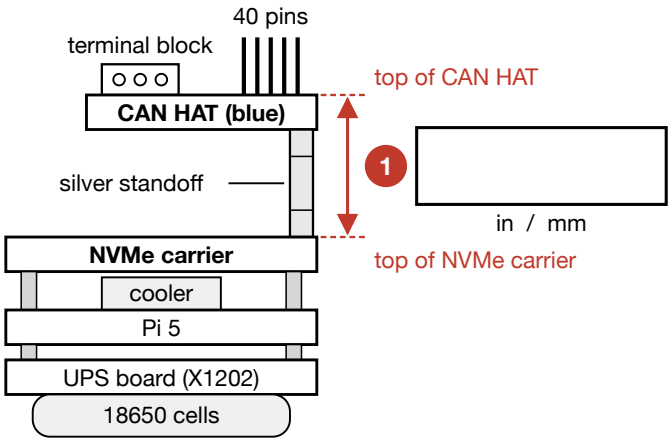
Nov 14, 2026

Write each value in the empty box next to its red number and circle **in** or **mm**. Send a photo of the sheet. Not to scale.

Dial caliper: one full turn of the needle = 1 inch. Each dial mark = 0.01 in. Whole inches come from the beam. Example: beam past 1, needle on 29 → **1.29 in**.

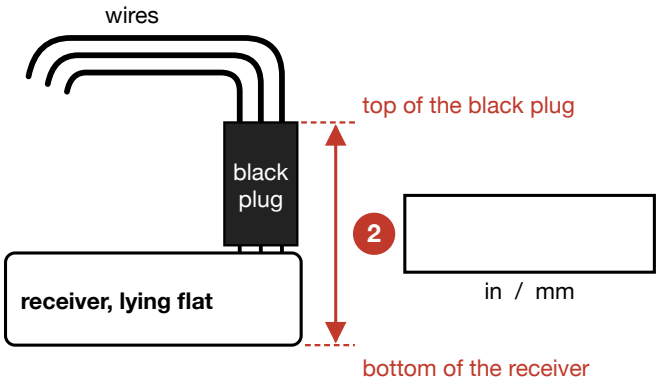
Already have, no need to redo: CAN HAT pin height ≈ 8 mm · Pi top to CAN HAT top 1.29 in · terminal block 0.355 in · receiver 1.30 × 0.785 × 0.40 in · receiver + lead on the Nano 5V pin · receiver header: male pins, S + –.

1. NVMe carrier to CAN HAT



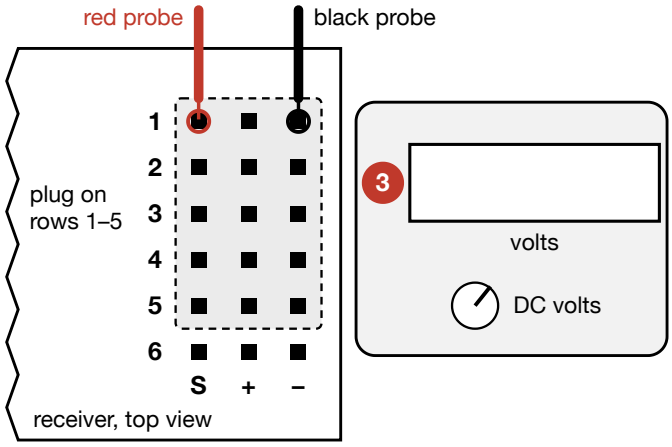
One jaw on the top face of the NVMe carrier, the other on the top face of the blue CAN HAT, right beside the silver standoff on the 40-pin side.

2. Receiver height with the plug on



Receiver flat, your harness plugged on. One jaw on the bottom of the receiver, the other on top of the black plug (not on the wires).

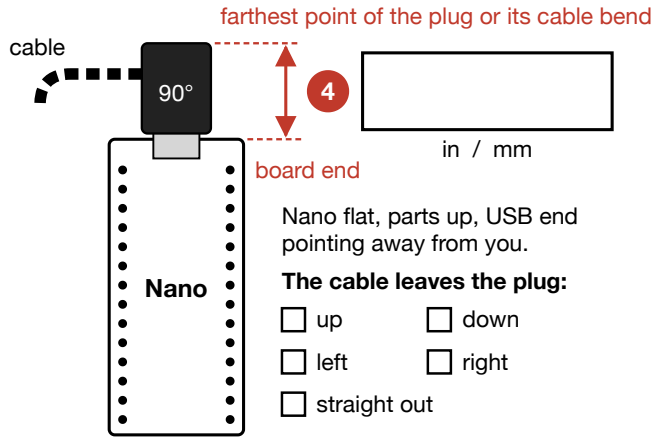
3. Receiver signal voltage (meter, not caliper)



Harness on the Nano, Nano on any USB port, transmitter on, stick centred, meter on DC volts. Red probe on row 1 S, black probe on row 1 –. Push the tips in from the top of the plug.

Blue LED on the receiver: ☐ on ☐ off

4. USB plug sticking out of the Nano



Nano flat, parts up, USB end pointing away from you.

The cable leaves the plug:

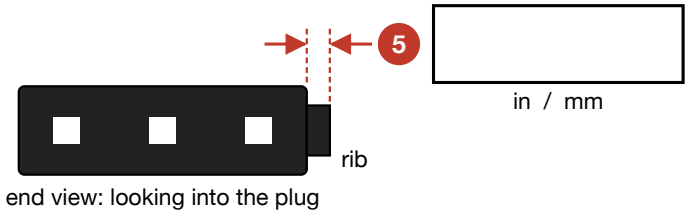
☐ up ☐ down

☐ left ☐ right

☐ straight out

Your 90° plug pushed fully in. One jaw on the USB end of the Nano's board, the other on the farthest point of the plug or its cable bend.

5. Rib on a servo plug (optional: skip if you have none)



Only if you have a servo lead whose plug (Futaba style) has a thin raised rib along one narrow side.

Jaws across the plug over the rib, then again just beside the rib. Write the difference: how far the rib stands out.